GitHub Actions & The GitHub ↔ AWS Cloud Bridge (Visual Architecture Guide)

How Code Moves from Local Git Push & Pull Requests ➔ Automated GitHub Actions Robots ➔ Live AWS Production Cloud (us-east-2)

Repo: cosmosanalytics/AWS

AWS Project: 966273137102 (us-east-2)

1. HOW CODE ENTERS GITHUB: GIT PUSH VS. PULL REQUEST

GIT WORKFLOW

git push (Direct Upload)

Uploads commits straight from your laptop to a GitHub branch (**git push origin main**).

- **Speed:** Instant — no waiting for human review.
- **When used:** Sole owner pushing directly to **main** or uploading to a sandbox branch.

S&OP Analogy: Directly updating the live plan as the system owner.

Pull Request / PR (Review Gate)

A formal request to merge a **feature-branch** into **main** after automated checks pass.

- **Safety:** Runs line-by-line AI diff review + CI tests *before* touching **main**.
- **Human Gate:** Pauses for explicit approval.

S&OP Analogy: Promoting a *Private Sandbox Scenario* to the *Committed Plan*.

2. WHAT IS GITHUB ACTIONS? (4 AUTOMATED ROBOT INSPECTORS ON UBUNTU-LATEST)

100% FREE ON PUBLIC REPOS

JOB 01 • AST AUDIT

AI Code Review

Boots a Linux VM & verifies all 3 Tiers and all 9 Bedrock models (Claude 4.6, DeepSeek, etc.) exist in **index.html**.

JOB 02 • DEVSECOPS

Zero-Leak Shield

Scans code to guarantee zero leaked AWS keys (**AKIA...**) and verifies the **X-Cosmos-Key** passcode gate.

JOB 03 • CI MATRIX

48/48 Unit Tests

Executes 48 automated assertions checking every canvas node, autonomy policy (**p1-p8**), and rollback rule.

JOB 04 • PRE-FLIGHT

AWS Gate Ping

Pings your live AWS **us-east-2** Lambda URL to confirm it is online and enforcing **401** security for strangers.

Why Is It Free? Public repos (cosmosanalytics/AWS) get UNLIMITED free GitHub Actions minutes! (Private repos: 2,000 free mins/mo, then $0.008/min)

3. HOW GITHUB & AWS CONNECT: EXACT DIRECTION OF INFORMATION FLOW (GITHUB ➔ AWS VS. AWS ➔ GITHUB)

SHA-256 OWNER PASSCODE GATE

GitHub Cloud — cosmosanalytics/AWS

1. Git Repository (main & gh-pages)
Stores index.html, pages.yml workflow, and commit history (#a1202bb).

2. GitHub Actions Runner VM
Runs Jobs 1-4 on every push & exposes run status via api.github.com.

3. GitHub Pages Web App (CDN)
Serves the interactive 3-Tier CI/CD Canvas at cosmosanalytics.github.io/AWS/.

① FORWARD TRIGGER (PUSH): GitHub Pages / Actions ➔ AWS Lambda

INFO FLOW: GITHUB ➔ AWS

Browser UI (unlocked with **cosmos2026**) or GitHub Actions Job 4 sends an encrypted **HTTPS POST** request with **X-Cosmos-Key**, selected model (**Claude Sonnet 4.6**), and audit prompt to your AWS Lambda Function URL in **us-east-2**.

② LIVE TELEMETRY FETCH (PULL): AWS Lambda Queries GitHub REST API

INFO FLOW: GITHUB ➔ AWS

AWS Lambda calls **GET https://api.github.com/repos/cosmosanalytics/AWS/commits** & **/actions/runs**. *(Even though AWS initiates the call, the data—live Commit SHA **#a1202bb** & Run **#39148195073**—flows **from GitHub into AWS**!)*

③ REVERSE RECEIPT & STATUS FLOW: AWS us-east-2 ➔ GitHub App & Repo

INFO FLOW: AWS ➔ GITHUB

AWS returns the live **Claude 4.6 Audit Text**, **Step Functions Execution ARN** (**exec-...**), and **S3 Artifact URI** back to the GitHub Pages Web UI (and can post a green ✅ **Deployed** commit status / PR comment back to the GitHub repo).

AWS Cloud (us-east-2) — Proj: 966273137102

1. AWS Lambda Orchestrator
cosmos-cicd-ai-orchestrator verifies SHA-256 passcode before any cloud spend.

2. Amazon Bedrock (9 Live FMs)
Claude Sonnet 4.6, Opus 4.6, Haiku 4.5, DeepSeek V3.2, Mistral Large 3, GPT-OSS 120B.

3. Step Functions & S3 Store
Runs cosmos-cicd-ai-pipeline-sfn & saves JSON receipt to s3://cosmos-ai-factory-cicd...

For Non-Owners (Public LinkedIn Visitors) — SIMULATED MODE ($0.00 Cost)

100% SAFE & FREE

Simulated GitHub: Plays back cached commit **#2588884** & Actions Run **#39148195073** locally without hitting GitHub API rate limits.

Simulated AWS: Plays back authentic pre-recorded receipts for all 9 Bedrock models locally in the browser (**$0.00** AWS bill).

Full UI Interactivity: Visitors can test all 3 Hierarchy Levels, Human Approval Gates, Canary Sliders, and Rollback Ledgers.

Server-Side Firewall: Unauthorized calls to the Lambda URL are blocked at the door with **HTTP 401 Unauthorized**.

For You (Owner Passcode Unlocked) — 100% REAL GITHUB + REAL AWS MODE

LIVE CLOUD EXECUTION

Real GitHub REST API: Pulls live commits, branch heads (**main/gh-pages**), and real GitHub Actions runs in real time.

Real Amazon Bedrock (us-east-2): Invokes live **Claude Sonnet 4.6** (or any of the 9 models) with the live GitHub commit context.

Real AWS Step Functions: Starts a new state machine execution (**exec-...**) stamped with the live GitHub SHA.

Real Amazon S3 Persistence: Writes the permanent JSON deployment & audit receipt to your S3 bucket in **us-east-2**.